VISHWAKARMA
AWARDS
A Maker Bhavan Foundation Initiative

MAKER
BHAVAN
FOUNDATION

VISHWAKARMA AWARDS 2026-27

VK26-IITI-HT-0087

Title

MEDICOOL

Portable Smart Thermoelectric Medicine Storage System.

Problem Context

Problem Statement:

People carrying temperature-sensitive medicines may struggle to maintain required storage conditions during travel, outdoor activities, power interruptions and in places without reliable refrigeration.

People carrying temperature-sensitive medicines may struggle to maintain required storage conditions during travel, outdoor activities, power interruptions and in places without reliable refrigeration.

Core innovation:

Rechargeable, sensor-controlled thermoelectric medicine storage

User Journey

Visual "Before State" Requirement

User relies on ice/gel packs, an insulated bag or nearby refrigeration. Cooling duration and temperature may require close attention during travel or outages.

Baseline temperature excursions, cooling duration and user burden have not yet been quantified; establish through user research and controlled comparison.

User Journey

Visual "After State" Requirement

Place medicine in the insulated chamber → sensor measures temperature → controller regulates the Peltier cooling stage → display/alerts report chamber conditions.

Performance gains are not yet measured. Planned metrics: time to reach target temperature, stability, recovery after opening and battery runtime.

MediCool | Problem Context & User Journey

Portable	Active	Sensor	Battery
MEDICINE STORAGE	THERMOELECTRIC	FEEDBACK CONTROL	RECHARGEABLE POWER



Figure 1: Insulin stored in pouch with ice bag

Target Beneficiaries & Current Challenge

Target users: people carrying temperature-sensitive medicines, including insulin users, travellers and people in areas with unreliable electricity.

Current options: include ice packs, insulated pouches or access to refrigeration. Suitability depends on the medicine, journey duration and ambient conditions.

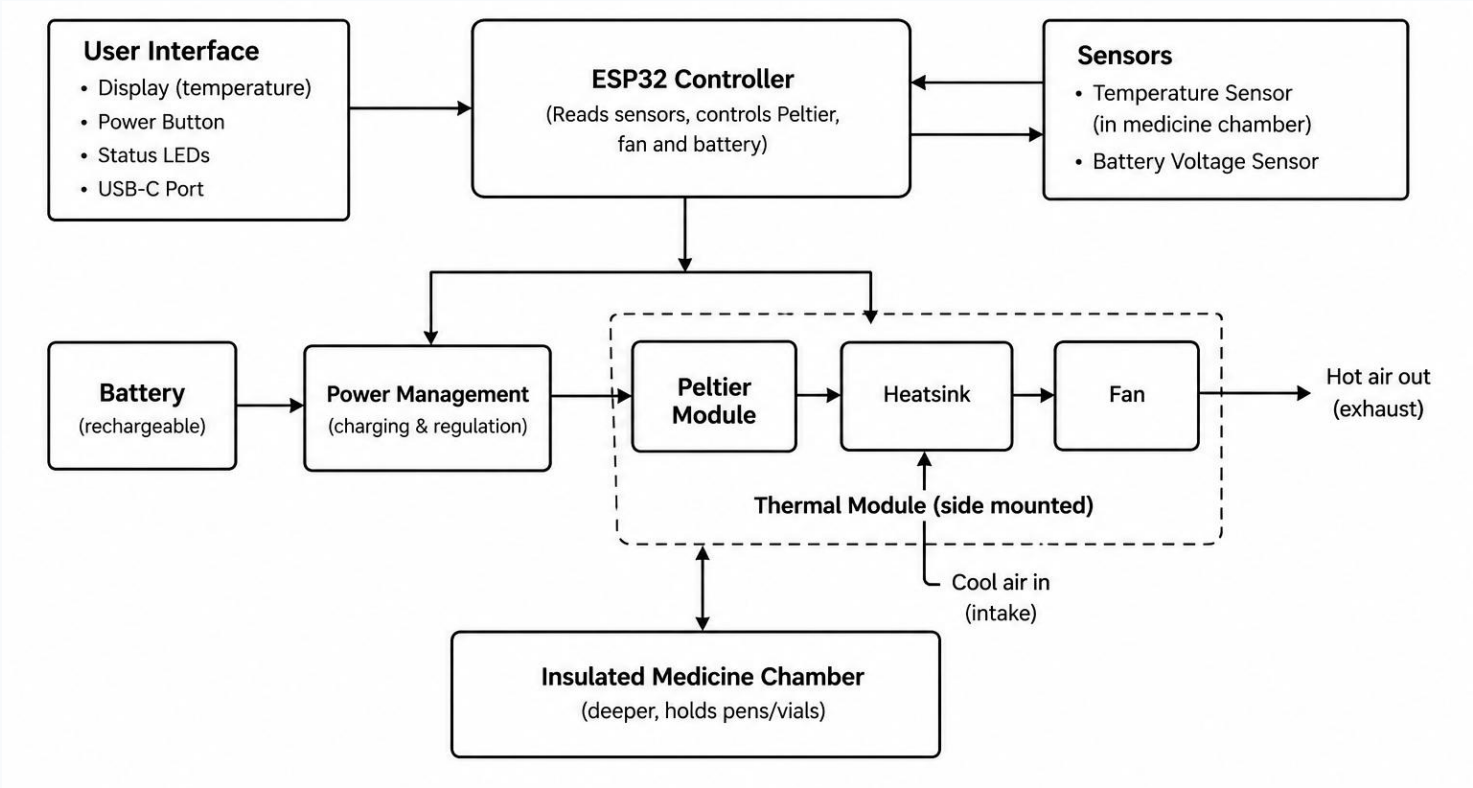
Current journey: Medicine requires controlled storage → user prepares passive cooling or finds refrigeration → temperature may change during travel or outages.

Evidence gap: Quantify the target users' needs and baseline temperature behaviour before claiming a measurable improvement.

Exemplar Submission Note: Team ID: VK26-IITI-HT-0087 | Track: Health Tech| Core Innovation: Portable Smart Thermoelectric Medicine Storage System.

System Architecture & Specs

System Block Diagram



System Architecture & Specs

Target Specifications

- Payload capacity: 1–2 insulin pens or equivalent vials
- Operating temperature: Target 2–8°C, subject to the specific insulin's storage requirements and validation
- Temperature monitoring: Real-time sensing with automatic cooling control
- Battery backup: Target 4 hours (to be validated)
- Power consumption: To be determined based on the selected Peltier module and cooling load
- Cooling method: Peltier thermoelectric cooling
- Charging: USB-C rechargeable battery
- Temperature alerts: High- and low-temperature warnings
- Portability: Compact, lightweight, backpack-friendly design
- Heat dissipation: Side-mounted heatsink and fan with a separate exhaust duct

System Architecture & Specs

Hardware Selection

Microcontroller	ESP32 — temperature monitoring and cooling control
Cooling unit	Peltier module — transfers heat away from the chamber
Temperature sensor	DS18B20 — measures chamber temperature
Heat dissipation	Heatsink + DC fan — removes heat from the Peltier hot side
Power supply	Rechargeable battery pack + BMS — portable power and battery protection
Charging	USB-C charging circuit — battery charging
Cooling control	MOSFET-based driver — switches the Peltier module
User interface	OLED display + power button and dedicated app — temperature and device status
Enclosure	3D-printed outer shell + insulation + inner liner — structure and thermal isolation

System Architecture & Specs



Figure: MediCool ,portable thermoelectric insulin storage system (concept image) .

Engineering Target Specifications & Subsystems

Parameter	Target Specification	Selected Hardware / Protocol
Storage capacity	1–2 insulin pens / equivalent vials	Insulated chamber with removable tray
Temperature control	Target 2–8°C, subject to insulin requirements	DS18B20 sensor + ESP32
Cooling system	Automatic cooling control	Peltier module + MOSFET driver
Battery backup	Target 4 hours	Rechargeable battery + BMS
Heat dissipation	Continuous hot-side heat removal	Heatsink + DC fan
User monitoring	Real-time temperature display	OLED display + ESP32

Specifications are preliminary design targets and require prototype validation.

Architecture Rationale: CAN-bus protocol was chosen over I2C/UART to eliminate high-frequency EMI noise generated by dual rotary tine motors under heavy torque loads.

CAD Design & Mechanical Specs

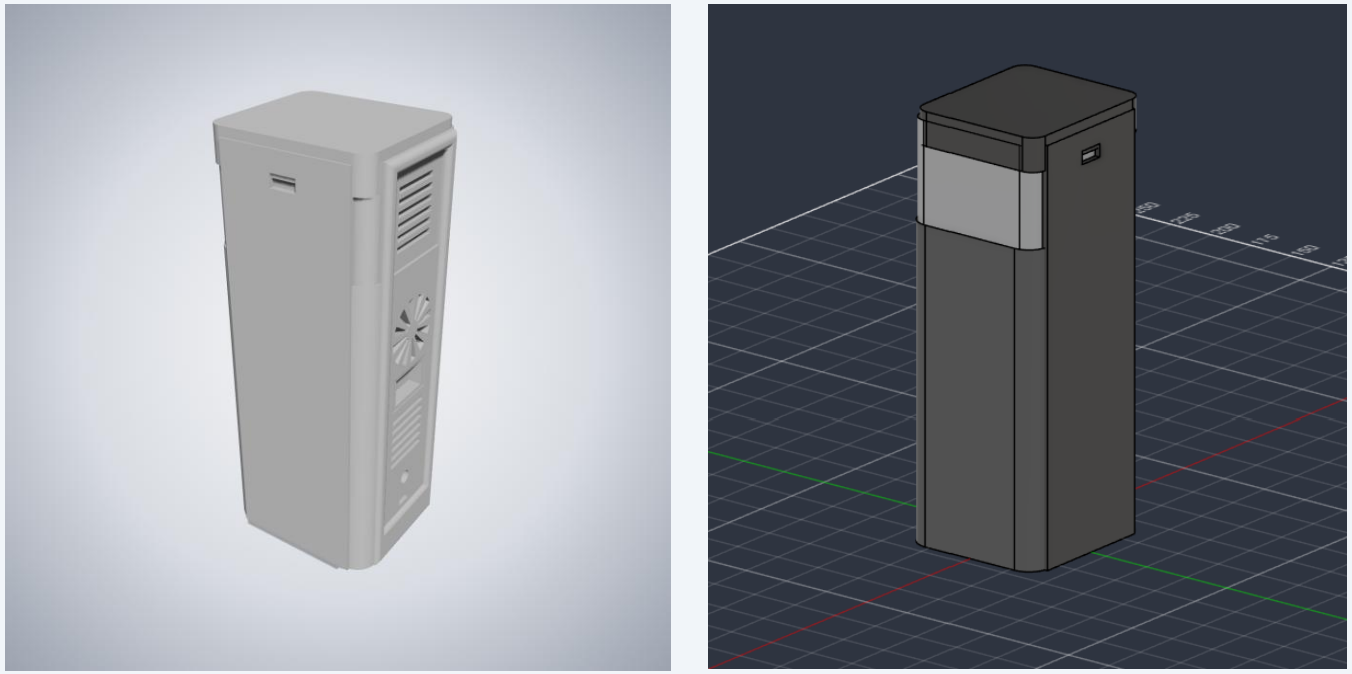


Figure: Current preliminary 3D model of MediCool. The design is subject to further refinement and modification.

Mechanical Design & Specs

- Enclosure: Compact cuboid structure
- Material: PETG casing proposed
- Storage: Insulated chamber with removable tray
- Cooling: Side-mounted Peltier assembly
- Airflow: Separate intake and exhaust vents
- Electronics: Dedicated lower compartment

Design Iteration: The modular layout separates the medicine chamber, cooling assembly and electronics to support compact packaging, heat dissipation and easier maintenance.

Electronics Schematics

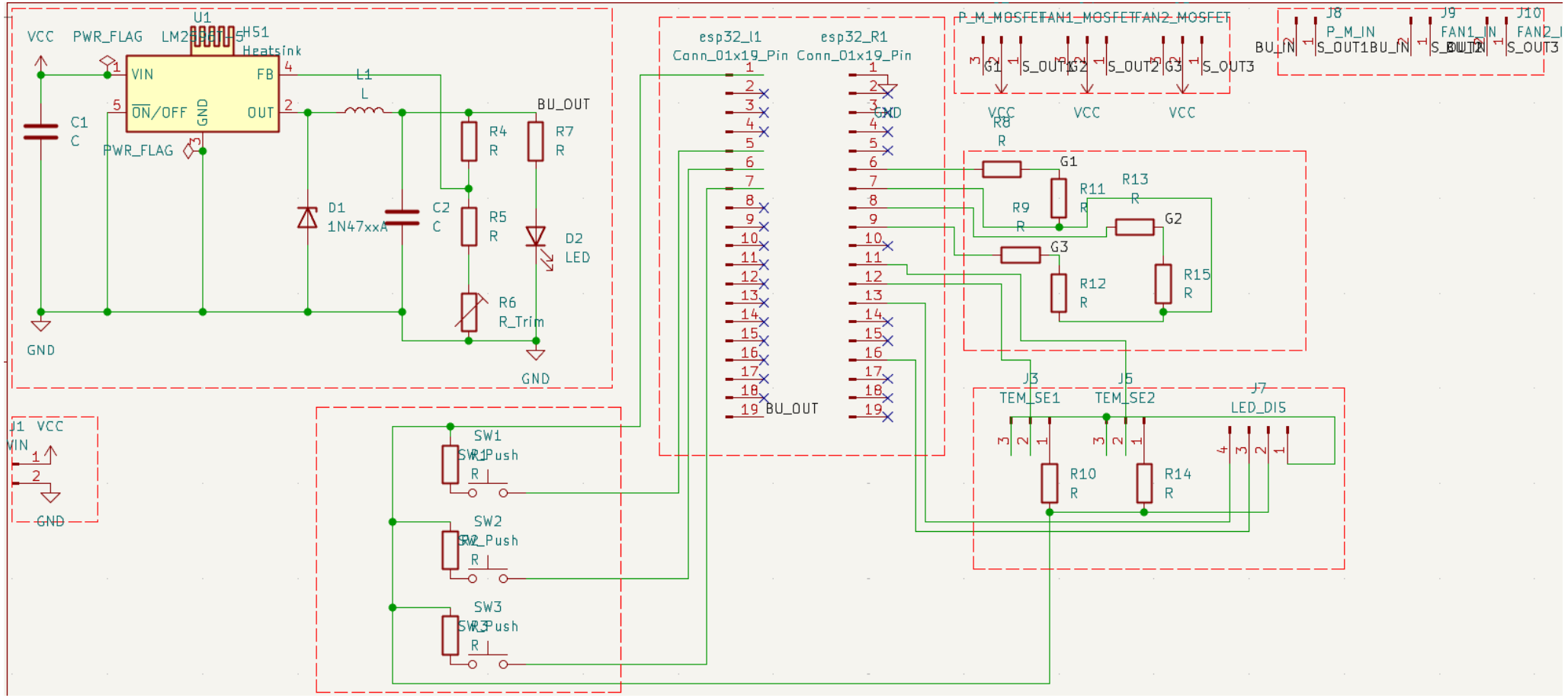


Figure: Electronics Schematics

BOM

Component	Part / Specification	Qty	Lead Time	Cost (INR)
ESP32 DevKit	ESP32-WROOM-32 development board	1	3–7 days	₹450
Temperature Sensor	DS18B20 waterproof digital probe	1	3–7 days	₹100
Peltier Module	TEC1-12706 thermoelectric cooler	1	3–7 days	₹350
MOSFET Driver	Logic-level MOSFET module	1	3–7 days	₹150
Heatsink & Fan	Aluminium heatsink with DC fan	1 set	3–7 days	₹350
Battery Pack	Rechargeable battery pack, capacity TBD	1	5–10 days	₹900
BMS	Compatible battery protection module	1	3–7 days	₹200

BOM

Charging Circuit	Compatible USB-C charging module	1	3–7 days	₹150
OLED Display	0.96-inch I²C OLED	1	3–7 days	₹180
Voltage Regulator	DC-DC buck converter	1	3–7 days	₹120
Passive Components	Resistors, capacitors, diodes and LEDs	1 lot	2–5 days	₹150
Switches	Momentary push buttons	3	2–5 days	₹30
Wiring & Connectors	Headers, terminals and hookup wires	1 lot	2–5 days	₹150
Insulated Enclosure	Outer case, insulation and removable tray	1 set	5–10 days	₹600
Prototype PCB	PCB or perfboard	1	3–7 days	₹200
Estimated Total				₹4,080

Maker Journey

Prototype Development and Engineering Progress

1. Concept Development

- Identified the challenge of transporting temperature-sensitive medicines when reliable refrigeration is unavailable.
- Proposed a portable, insulated medicine chamber using thermoelectric cooling and temperature feedback.

2. Electronics Development

- Designed the controller, temperature-sensing, power-regulation and cooling-driver circuits.
- Prepared the circuit schematic and preliminary Bill of Materials.
- Selected the ESP32, DS18B20 sensor, Peltier module, heatsink, fan and rechargeable power system as the proposed core hardware.

3. Mechanical Development

- Planned a compact enclosure with an insulated medicine chamber and a separate hot-side cooling assembly.
- Considered serviceability, heat removal, wiring access and portability in the layout.

4. Next Engineering Milestones

- Complete the verified circuit and enclosure design.
- Assemble the electronics and integrate the thermal module.
- Test temperature regulation, power consumption, battery runtime and recovery after opening the chamber.

Maker Journey

Work-In-Progress

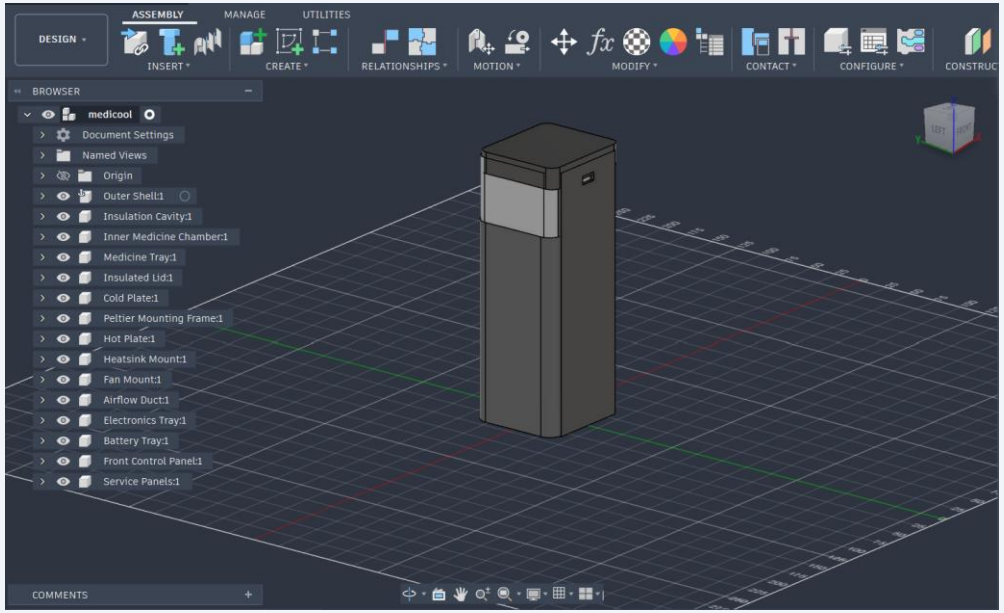


Figure: CAD design environment showing the development of the proposed system.

Raw Repository Linkage

https://github.com/pranaykpradeep/medicool

Institutional Sign-Off: Work audited and certified by Faculty Guide Prof. S. K. Kulkarni, Dept. of Mechatronics, COEP Technological University Pune.

Safety & Hazard Mitigation

Electrical Safety

- Use a correctly rated battery protection system, fuse and insulated wiring.
- Protect against short circuits, overcurrent, reverse polarity and accidental contact with live conductors.
- Use a charging circuit compatible with the selected battery chemistry.

Medicine Protection

Define temperature limits according to the specific medicine's storage instructions.
Prevent the Peltier cold surface from directly contacting stored medicines.
Validate temperature uniformity inside the chamber before claiming safe storage performance.

Thermal Safety

- Monitor the medicine chamber temperature continuously.
- Provide high- and low-temperature alerts when the chamber exceeds configured limits.
- Use a heatsink and fan to remove heat from the Peltier hot side.
- Stop or limit cooling if the hot-side temperature, current or other monitored parameters exceed validated safety limits.

Roadmap

Month 1 — Design Finalisation

Finalise circuit schematic, component specifications and BOM.
Complete enclosure CAD and insulation layout.
Calculate the cooling load, battery capacity and heat-dissipation requirements.
Procure the essential components.

Month 2 — Fabrication and Assembly

Assemble the ESP32 control and temperature-sensing circuit.
Integrate the Peltier module, heatsink and fan.
Fabricate the insulated enclosure and install the battery and charging system.
Perform electrical continuity and power-up checks.

Month 3 — Firmware and System Integration

Implement temperature monitoring and automatic cooling control.
Integrate the OLED display and temperature alerts.
Test sensor readings, cooling response and power consumption.
Correct wiring, thermal and firmware issues identified during testing.

Month 4 — Testing and Demonstration

Measure time to reach the target temperature.
Evaluate temperature stability and recovery after opening the chamber.
Measure battery runtime under defined test conditions.
Document results, complete safety checks and prepare the Stage 3 demonstration.

Safety by Design

Failure mode	Potential consequence	Mitigation
Temperature sensor failure	Incorrect temperature measurement	Validate readings and detect sensor disconnection or invalid data.
Peltier overheating	Reduced cooling performance or component damage	Use a properly rated heatsink, fan and thermal shutdown logic.
Battery undervoltage	Unexpected system shutdown	Monitor battery voltage and provide a low-battery warning.
Short circuit or excessive current	Wiring or component damage	Add suitable fuse protection and verify current ratings.
Poor chamber insulation	Excessive power consumption and unstable temperature	Seal thermal gaps and evaluate insulation performance.
Temperature exceeds limits	Medicine may be exposed to unsuitable conditions	Trigger an alert and record the temperature excursion where supported.
Failure mode	Potential consequence	Mitigation
Temperature sensor failure	Incorrect temperature measurement	Validate readings and detect sensor disconnection or invalid data.
Peltier overheating	Reduced cooling performance or component damage	Use a properly rated heatsink, fan and thermal shutdown logic.
Battery undervoltage	Unexpected system shutdown	Monitor battery voltage and provide a low-battery warning.

Technical Challenges & Risks

Thermal Management

Maintaining the target 2–8°C chamber temperature under varying ambient conditions.
Preventing heat accumulation on the Peltier module's hot side.

Battery Runtime

Balancing Peltier power consumption with the target 4-hour battery backup.
Managing battery capacity, peak current and overall device weight.

Temperature Stability

Preventing temperature fluctuations when the lid is opened.
Ensuring uniform cooling throughout the medicine chamber.

Electronics Integration

Integrating the ESP32, DS18B20 sensor, MOSFET driver and power-management circuit.
Protecting against sensor failure, overcurrent and battery undervoltage.

Key Technical Risks

Insufficient cooling: Inadequate heatsink capacity or poor insulation.
Short battery life: Higher-than-expected cooling power demand.
Temperature excursion: Sensor faults or delayed cooling response.
Component failure: Excessive current or inadequate electrical protection.

Potential Solutions & Risk Mitigation (1 Slide)

Temperature Control

Risk: Temperature may fluctuate or fall below the required range.
Mitigation: Use sensor-based feedback control, temperature alerts and independent temperature monitoring during testing.

Power & Heat Dissipation

Risk: High power consumption, battery depletion or overheating on the hot side.
Mitigation: Select a suitably rated battery with protection circuitry, and use a heatsink, fan and separated airflow path.

Sensor or System Failure

Risk: Sensor faults or controller failure may interrupt cooling.
Mitigation: Detect invalid sensor readings, trigger fault alerts and test safe shutdown or fallback behaviour.

Medicine Safety

Risk: Poor insulation, condensation or accidental freezing may affect storage conditions.
Mitigation: Use an insulated, sealed chamber and validate temperature uniformity, condensation control and low-temperature cut-off.

Example: Technical Challenges & Risks

Bottleneck: Wet Soil Clogging

Observed Risk: High-moisture (>25%) black cotton soil adheres to rotary cultivator teeth, causing rotational binding within 12 minutes of continuous furrow tilling.
Mitigation: Installed stationary low-friction PTFE scraper blades offset 1.5mm from the tine arc to dynamically peel away clay buildup on every rotation.

Bottleneck: Thermal Throttling

Observed Risk: Hub motor temperatures reached 68°C in simulation during sustained noon operations at 38°C ambient Maharashtra field heat.
Mitigation: Retrofitted extruded radial aluminum heatsinks directly coupled via 3.0 W/mK silicone thermal pads, stabilizing motor casing temps below 47°C.

Standards & Enclosure Specs

ISO 4254-1 Compliance: 2mm sheet steel blade shrouds eliminate projectile pebble hazards.
IP65 Ingress Sealing: Gasket-sealed ASA electronics bay with IP68 PG7 cable glands tested against direct pressurized spray.
IEC 62133: BMS equipped with cell-level thermal sensors and overcharge isolation.

Empirical Verification: Benchtop clay test rig showed a 94% reduction in mechanical rotational drag after installing the PTFE scraper combs.

Image Sources



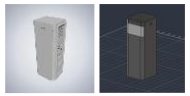
https://www.flipkart.com/outer-woods-insulin-cooler-travel-bag-diabetics-two-ice-gel-packs-pack/p/itm0fca7d4816eb5

Source: www.flipkart.com

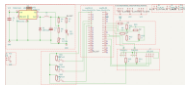



Ai concept image.

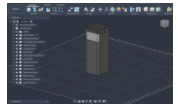
Source: chatgpt



3d design .



Circuit diagram



3d designing interface with work tree